RAJIV GANDHI INSTITUTE OF TECHNOLOGY KOTTAYAM

DEPARTMENT OF MECHANICAL ENGINEERING

M.Tech Data Science

PrivGraphAI: A Privacy-Preserving On-Device Agentic GraphRAG Framework for Multimodal Document Question Answering

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Abstract: Retrieval-Augmented Generation (RAG) enables effective domain-specific question answering, but cloud deployments expose sensitive documents to privacy risk, recurring cost, and infrastructure dependence. Building on the on-device foundation established by PrivSumAI, this work proposes PrivGraphAI, a fully on-device Agentic GraphRAG framework for privacy-preserving, multimodal document question answering over text, tables, figures, and formulas. It integrates Docling for structural extraction, quantized language and vision-language models served locally via Ollama, and LangGraph for stateful agent orchestration. Specialized extraction agents build a document-level knowledge graph; a routing agent selects the retrieval path, while a verification agent triggers re-retrieval when evidence is insufficient. PrivGraphAI will be evaluated against an on-device Vector RAG baseline and a cloud-hosted GraphRAG baseline on a shared document and question set, measuring answer correctness, groundedness, retrieval precision, latency, RAM utilization, throughput, and graph-construction time on consumer-grade hardware without a dedicated GPU.

Problem: Existing on-device RAG systems rely on flat vector retrieval over plain text, struggling with tables, figures, formulas, and multi-hop queries. Graph-based reasoning solves this in the cloud, but its cost on constrained local hardware remains uncharacterized.

Objectives:

1. Design and implement PrivGraphAI, integrating Docling, quantized models served via Ollama, and LangGraph agent orchestration for on-device multimodal processing.
2. Construct a multimodal knowledge graph via specialized extraction agents, with routing and evidence-verification agents enabling reliable multi-hop reasoning.
3. Establish on-device Vector RAG and cloud-hosted GraphRAG baselines for controlled comparison.
4. Empirically characterize trade-offs among privacy, answer quality, reasoning capability, and computational efficiency on consumer-grade hardware.

Progress: Completed the literature review; finalized the research gap, objectives, and abstract; and prepared the architecture design and Zeroth Review presentation. Next: implement the Docling–Ollama–LangGraph pipeline, build both baselines, and begin comparative evaluation.

References:

1. D. Edge et al., "From Local to Global: A Graph RAG Approach to Query-Focused Summarization," arXiv:2404.16130, 2024. (Base paper)
2. P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," NeurIPS, 2020.
3. C. Auer et al., "Docling Technical Report," arXiv:2408.09869, IBM Research, 2024.

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